

EEE4114F

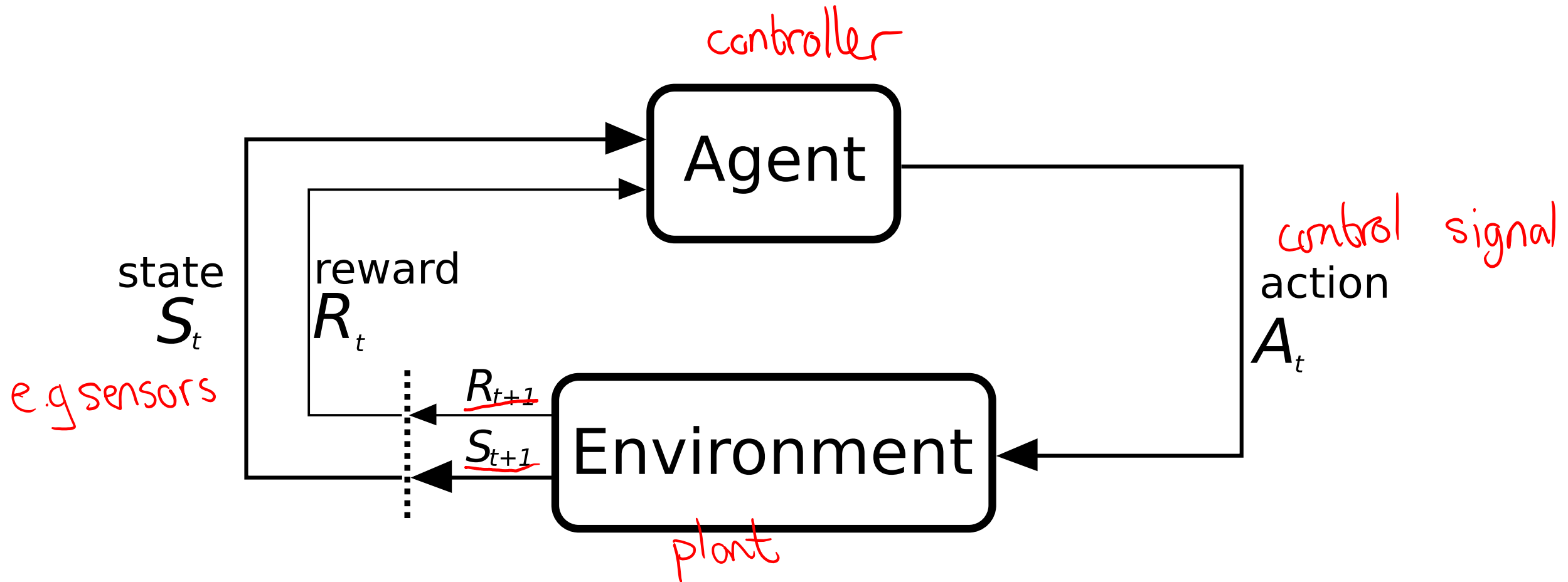
Finite Markov Decision Processes

Markov Decision Process

- Formalization of sequential decision making
- Actions influence
 - Immediate rewards
 - Future states
 - Future rewards



The Agent-environment Interface



MDP Dynamics

- At each time step, t
 - Agent receives state information, $S_t \in \mathcal{S}$
 - Based on state information selections an action, $A_t \in \mathcal{A}$
 - Agent receives reward, $R_{t+1} \in \mathcal{R}$
 - New state, $S_{t+1} \in \mathcal{S}$
- $p(s', r \mid s, a) \doteq \Pr\{S_t = s', R_t = r \mid \underline{S_{t-1}} = s, \underline{A_{t-1}} = a\}$
 - For all $s', s \in \mathcal{S}, \underline{r} \in \mathcal{R}, \underline{a} \in \mathcal{A}$
 - Probability of each possible S_t and R_t is only dependant on the previous state and action

Goals and Rewards

- Reward, $R_{t+1} \in \mathcal{R}$
 - Numerical value
 - Signal from the environment
- The goal for in RL is to maximize expected value of cumulative sum of a scalar signal (reward)
- Reward should relate to final goal for the task, not subgoals
- E.g. Simple game
 - +1 reward for winning
 - -1 reward for losing
 - 0 reward for doing something not too interesting
- E.g Chip design
 - $r = -\text{wirelength} - \lambda \times \text{congestion} - \gamma \times \text{density}$

Returns and Episodes

- Return

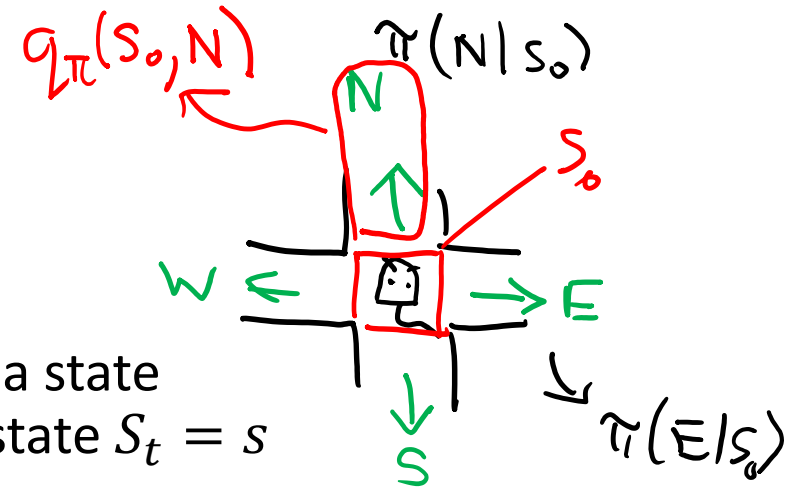
- $G_t \doteq R_{t+1} + R_{t+2} + R_{t+3} + \dots + R_T$
- Where T is a final time step
- This is an episodic task as there is a defined end to the “episode”
- The final state is called the terminal state
- System is reset before starting another episode (independent)
- If $T = \infty$, then it is a continuing task e.g. process control

- Discounting

- $G_t \doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$
 - $0 \leq \gamma \leq 1$, γ is known as the discount rate
 - Determines present value of future rewards
 - $\gamma = 0$, implies you only care about immediate reward
 - As γ approaches 1 future rewards have more impact

Policies and Value Functions

- Policy, $\pi(a|s)$
 - Dictates what actions and agent is going to take based on a state
 - Probability that the action is $A_t = a$ given the agent is in state $S_t = s$
- Value functions, $v_\pi(s)$ or $q_\pi(s, a)$:
 - $v_\pi(s) \doteq \mathbb{E}_\pi[G_t | S_t = s] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s\right], \forall s \in \mathcal{S}$
 - $\mathbb{E}[X] = \sum_{i=1}^k x_i p_i$
 - state-value function
 - How good is it to be in a particular state, s .
 - Start in, s , and follow policy, π , thereafter
 - $q_\pi(s, a) \doteq \mathbb{E}_\pi[G_t | S_t = s, A_t = a] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, A_t = a\right]$
 - action-value function
 - How good is it to take an action, a , in state, s .
 - Start in, s , take action, a , and follow policy, π



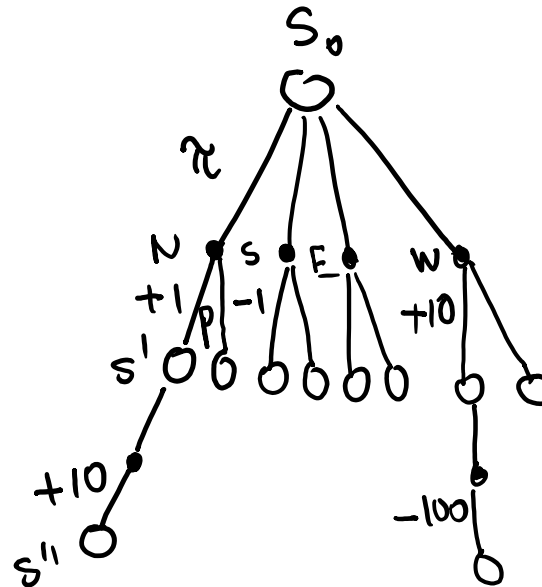
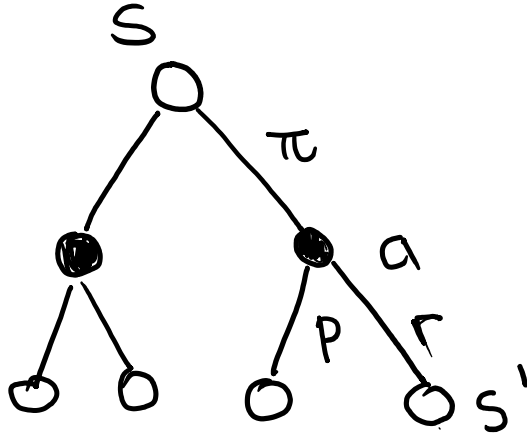
Value function cont.

$$\mathbb{E}[R_{t+1} + \gamma G_{t+1}]$$

$$\mathbb{E}[x] = \sum x_i p_i$$

dynamics

- $v_\pi(s) \doteq \mathbb{E}_\pi[G_t | S_t = \underline{s}] = \sum_a \underbrace{\pi(a|s)}_{\text{policy}} \sum_{s', r} p(s', r | s, a) [\underbrace{r}_{\text{dynamics}} + \gamma v_\pi(s')]$
 - Bellman equation for v_π
- Described graphically with a backup diagram



Example

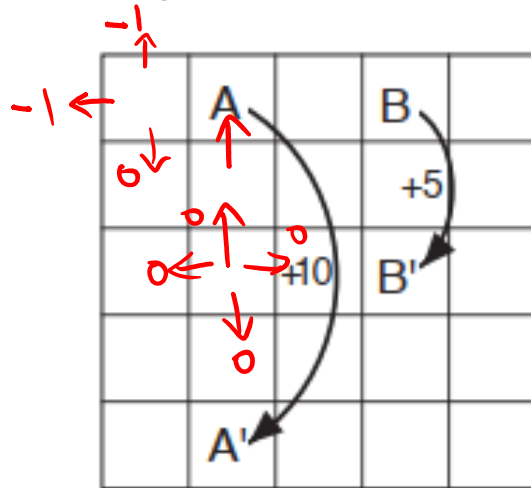
- Gridworld

- 4 actions: N, S, E, W

- taken with **equal** probability $\pi(N|s) = \frac{1}{4}$; $\pi(S|s) = \frac{1}{4}$...

- move **deterministically** one cell in that direction

- $\gamma = 0.9$



3.3	<u>8.8</u>	4.4	<u>5.3</u>	1.5
1.5	3.0	2.3	1.9	0.5
0.1	<u>0.7</u>	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

V_π

Let's verify the calculated values

	3.0	2.3	1.9
s_6	s_7	s_8	
0.7	0.7	0.4	
s_3	s_4	s_5	
-0.4	-0.4	-0.6	
s_0	s_1	s_2	

v_π

$$v_\pi = \mathbb{E}[G_t | s_t = s]$$

$$= \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_\pi(s')]$$

$$\pi(a|s) = \frac{1}{4}$$

$$p(s', r | s, a) = \begin{cases} 1 & \text{if } s = s_1, s_3, s_5, 7 \\ 0 & \text{if } s = s_0, s_2, s_6, s_8 \end{cases}$$

$$v_\pi(s_4) = \frac{1}{4} [0.9(2.3 + 0.4 + (-0.4)) + 0.7]$$

$$= \underline{0.675}$$

Optimal Policies and Value Functions

- The policy with a greater expected return is the better policy
- Always at least one policy – the optimal policy – that is $\geq$ all others
 - Denoted by, π_*
- The optimal state-value function
 - $v_* \doteq \max_{\pi} v_{\pi}(s)$
- The optimal action-value function
 - $q_* \doteq \max_{\pi} q_{\pi}(s, a)$

Next up:

Temporal Difference Learning